MAS 812: Nonparametric Methods

Lecture 3 & 4: Two-Sample Nonparametric Tests

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Learning Objectives

By the end of this lecture, you will be able to:

- Conduct and interpret the Mann-Whitney U test (Wilcoxon rank-sum test)
- Apply the median test for comparing two independent samples
- Perform the Kolmogorov-Smirnov two-sample test
- Understand permutation tests for two-sample problems
- Estimate the shift parameter between two populations
- Apply the Wald-Wolfowitz runs test for homogeneity
- Discuss power and efficiency of two-sample nonparametric tests

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1 Introduction to Two-Sample Problems

In two-sample problems, we have independent random samples from two populations and wish to test whether the populations differ in location, scale, or overall distribution.

Let $X_1, X_2, \ldots, X_m$ be a random sample from population 1 with continuous distribution function F, and $Y_1, Y_2, \ldots, Y_n$ be a random sample from population 2 with continuous distribution function G. We assume the samples are independent.

Common Hypotheses to test include:

- Location shift: $G(x) = F(x - \Delta)$ for some shift parameter Δ
- Stochastic dominance: $F(x) \leq G(x)$ for all x (or vice versa)
- Equality of distributions: $F = G$

Parametric Approach: The Two-Sample t-test

The classical parametric approach assumes F and G are normal with equal variances:

$$t = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2} \text{ under } H_0$$

However, this test is sensitive to non-normality, unequal variances, and outliers.

2 The Wilcoxon Rank-Sum Test / Mann-Whitney U Test

The Wilcoxon rank-sum test (also called the Mann-Whitney U test) is the most popular nonparametric alternative to the two-sample t-test. It tests whether one population tends to produce larger values than the other.

Assumptions

1. The two samples are *independent* random samples from *continuous* distributions
2. The measurement *scale is at least ordinal*
3. Under H_0, the two *distributions are identical* (for the test of location shift, we assume identical shapes)

2.1 Test Procedure (Wilcoxon Rank-Sum Form)

1. Combine the two samples into one group of size $N = m + n$
2. Rank all observations from smallest (rank 1) to largest (rank N), handling ties by assigning average ranks
3. Let W be the sum of ranks from the first sample (or the smaller sample)
4. Under H_0, the ranks are randomly distributed between the two samples

2.2 Test Procedure (Mann-Whitney U Form)

The Mann-Whitney U statistic is equivalent to the Wilcoxon rank-sum statistic:

$$U = mn + \frac{m(m+1)}{2} - W$$

where W is the rank sum for the first sample. Alternatively,

$$U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(X_i > Y_j)$$

which counts the number of times an X observation exceeds a Y observation.

Relationship Between W and U

$$W = U + \frac{m(m+1)}{2}$$

$$U = mn - U' \quad \text{where } U' = \sum_{i=1}^{m} \sum_{j=1}^{n} I(X_i < Y_j)$$

Distribution Under Null Hypothesis

Under H_0 that the two distributions are identical, the distribution of W (or U) is symmetric with:

$$\mathbb{E}[W] = \frac{m(m+n+1)}{2}, \quad \text{Var}(W) = \frac{mn(m+n+1)}{12}$$

$$\mathbb{E}[U] = \frac{mn}{2}, \quad \text{Var}(U) = \frac{mn(m+n+1)}{12}$$

For large samples ($m, n \geq 10$), the normal approximation can be used:

$$Z = \frac{U - mn/2}{\sqrt{mn(m+n+1)/12}} \sim N(0,1)$$

Handling Ties

When ties occur, assign average ranks and adjust the variance:

$$\mathrm{Var}(U) = \frac{mn}{12}\left[(m+n+1) - \frac{\sum_{j=1}^{g}(t_j^3 - t_j)}{(m+n)(m+n-1)}\right]$$

where g is the number of tied groups and t_j is the size of the j-th tied group.

Example 2.1 (Wilcoxon Rank-Sum Test)**.** A researcher wants to compare the effectiveness of two pain relievers. Patients are randomly assigned to either Drug A or Drug B, and their pain relief times (in minutes) are recorded:

Drug A: 20, 25, 22, 28, 24, 26, 23, 27 Drug B: 19, 21, 24, 18, 22, 20, 23, 19, 20

Test at $\alpha = 0.05$ whether the two drugs differ in pain relief time.

Solution:

Step 1: State hypotheses

$$H_0 : \text{The two distributions are identical} \quad \text{vs.} \quad H_1 : \text{The distributions differ}$$

Step 2: Combine and rank all observations

Observation	Value	Group	Rank
1	18	B	1
2	19	B	2.5
3	19	B	2.5
4	20	A	5
5	20	B	5
6	20	B	5
7	21	B	7
8	22	A	8.5
9	22	B	8.5
10	23	A	10.5
11	23	B	10.5
12	24	A	12.5
13	24	B	12.5
14	25	A	14
15	26	A	15
16	27	A	16
17	28	A	17

Note: $m = 8$ (Drug A), $n = 9$ (Drug B), $N = 17$

Step 3: Calculate rank sum for Drug A

$$W_A = 5 + 8.5 + 10.5 + 12.5 + 14 + 15 + 16 + 17 = 98.5$$

Step 4: Calculate expected value and variance under H_0

$$\mathbb{E}[W_A] = \frac{m(m+n+1)}{2} = \frac{8(8+9+1)}{2} = \frac{8 \times 18}{2} = 72$$

$$\text{Var}(W_A) = \frac{mn(m+n+1)}{12} = \frac{8 \times 9 \times 18}{12} = \frac{1296}{12} = 108$$

$$SD = \sqrt{108} = 10.39$$

Step 5: Calculate U statistic

$$U = W_A - \frac{m(m+1)}{2} = 98.5 - \frac{8 \times 9}{2} = 98.5 - 36 = 62.5$$

Step 6: Normal approximation

$$Z = \frac{U - mn/2}{\sqrt{mn(m+n+1)/12}} = \frac{62.5 - 36}{\sqrt{108}} = \frac{26.5}{10.39} = 2.55$$

Two-sided p-value $\approx 2 \times \mathbb{P}(Z > 2.55) = 2 \times 0.0054 = 0.0108$

Step 7: Conclusion Since p-value $= 0.0108 < 0.05$, we reject H_0. There is significant evidence that the pain relief times differ between the two drugs. The higher rank sum for Drug A suggests it tends to have longer pain relief times.

3 Estimation of Shift Parameter

The shift parameter Δ represents the amount by which one population is shifted relative to the other: $G(x) = F(x - \Delta)$.

Hodges-Lehmann Estimator for Two Samples

The Hodges-Lehmann estimator of the shift parameter Δ is the median of all possible differences between an observation from sample 1 and an observation from sample 2:

$$\hat{\Delta}_{HL} = \text{median}\{X_i - Y_j : i = 1, \ldots, m, j = 1, \ldots, n\}$$

Properties

- Robust estimator of location shift
- Consistent and asymptotically normal
- Approximately 95.5% efficient relative to the difference in means when data are normal
- More efficient for heavy-tailed distributions

Example 3.1 (Hodges-Lehmann Estimator for Two Samples)**.** For the pain relief data from the previous example, estimate the shift parameter Δ (how much longer Drug A lasts compared to Drug B).

Solution:

Drug A: 20, 25, 22, 28, 24, 26, 23, 27 Drug B: 19, 21, 24, 18, 22, 20, 23, 19, 20

Compute all $8 \times 9 = 72$ differences $X_i - Y_j$:

$X_i \backslash Y_j$	19	21	24	18	22	20	23	19	20
20	1	-1	-4	2	-2	0	-3	1	0
25	6	4	1	7	3	5	2	6	5
22	3	1	-2	4	0	2	-1	3	2
28	9	7	4	10	6	8	5	9	8
24	5	3	0	6	2	4	1	5	4
26	7	5	2	8	4	6	3	7	6
23	4	2	-1	5	1	3	0	4	3
27	8	6	3	9	5	7	4	8	7

Sorted differences (72 values): -4, -4, -3, -3, -2, -2, -2, -1, -1, -1, -1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 9, 10

Median of 72 values = average of 36th and 37th values: 36th value = 3, 37th value = 3

$\hat{\Delta}_{HL} = (3 + 3)/2 = 3$ minutes

Interpretation: The estimated shift is 3 minutes, meaning Drug A tends to provide pain relief for about 3 minutes longer than Drug B.

3.1 Confidence Interval for Shift Parameter

A confidence interval for the shift parameter Δ can be obtained by inverting the Wilcoxon rank-sum test.

Procedure

Let $D_{(1)} \leq D_{(2)} \leq \cdots \leq D_{(mn)}$ be the ordered differences $X_i - Y_j$. A $100(1-\alpha)\%$ confidence interval for Δ is:

$$(D_{(k)}, D_{(mn+1-k)})$$

where k is chosen so that $\mathbb{P}(W < k) \leq \alpha/2$ under the null distribution, with W being the Wilcoxon rank-sum statistic.

For large samples, k can be approximated by:

$$k \approx \frac{mn}{2} - z_{\alpha/2}\sqrt{\frac{mn(m+n+1)}{12}}$$

Example 3.2 (Confidence Interval for Shift Parameter)**.** For the pain relief data, construct a 95% confidence interval for the shift parameter.

Solution:

From the previous example, we have $m = 8$, $n = 9$, $mn = 72$.

For a 95% CI, we need k such that $\mathbb{P}(W < k) \leq 0.025$. Using the normal approximation:

$$k \approx \frac{mn}{2} - z_{0.025}\sqrt{\frac{mn(m+n+1)}{12}} = 36 - 1.96\sqrt{\frac{72 \times 18}{12}}$$

$$= 36 - 1.96\sqrt{108} = 36 - 1.96 \times 10.39 = 36 - 20.36 = 15.64$$

Rounding to nearest integer, $k = 16$ (conservative choice).

Thus, the 95% CI uses the 16th smallest and 16th largest differences.

From the sorted differences:

- 16th smallest = 1
- 16th largest = position $72 - 16 + 1 = 57$, which is 6

95% CI for Δ: (1, 6) minutes.

Interpretation: We are 95% confident that Drug A provides pain relief for 1 to 6 minutes longer than Drug B.

4 The Median Test (Mood's Median Test)

The median test is a simple nonparametric test for comparing the medians of two independent samples. It is less powerful than the Wilcoxon test but has fewer assumptions.

Assumptions

- Two independent random samples
- Measurement scale at least ordinal

Test Procedure

1. Combine the two samples and find the overall median M of the combined data
2. Create a 2×2 contingency table:

	Sample 1	Sample 2	Total
Above median	a	b	$a+b$
Below median	c	d	$c+d$
Total	$a+c$	$b+d$	N

3. Under H_0 that the populations have the same median, the test statistic is:

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

which follows a χ^2 distribution with 1 degree of freedom (with Yates' continuity correction for small samples)

Example 4.1 (Median Test)**.** Using the pain relief data, test whether the two drugs have the same median relief time.

Solution:

Step 1: Combined data: Drug A: 20, 25, 22, 28, 24, 26, 23, 27 Drug B: 19, 21, 24, 18, 22, 20, 23, 19, 20

Combined sorted: 18, 19, 19, 20, 20, 20, 21, 22, 22, 23, 23, 24, 24, 25, 26, 27, 28

Overall median = 23 (9th value)

Step 2: Create contingency table:

Values above 23:
Drug A: 25, 28, 24, 26, 27 (5 values)
Drug B: 24 (1 value)

Values below 23:
Drug A: 20, 22 (2 values)
Drug B: 18, 19, 19, 20, 20, 21, 22 (7 values)

Values equal to 23: Drug A has 1, Drug B has 1 (these can be excluded or split)

Contingency table (excluding ties):

	Drug A	Drug B	Total
Above median	5	1	6
Below median	2	7	9
Total	7	8	15

Step 3: Calculate chi-square statistic

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} = \frac{15(5\times 7 - 1\times 2)^2}{6\times 9\times 7\times 8}$$

$$= \frac{15(35-2)^2}{3024} = \frac{15\times 33^2}{3024} = \frac{15\times 1089}{3024} = \frac{16335}{3024} = 5.40$$

Step 4: Compare to χ^2_1 critical value At $\alpha = 0.05$, $\chi^2_{1,0.05} = 3.84$

Since $5.40 > 3.84$, we reject H_0.

Step 5: Conclusion There is significant evidence that the two drugs have different medians. Drug A tends to have higher pain relief times.

5 Kolmogorov-Smirnov Two-Sample Test

The Kolmogorov-Smirnov two-sample test is a general test for equality of two continuous distributions. It is sensitive to differences in location, scale, and shape.

Assumptions

- Two independent random samples from continuous distributions
- The test is consistent against all types of differences between distributions

5.1 Test Procedure

Let $F_m(x)$ be the empirical CDF of sample 1 and $G_n(x)$ be the empirical CDF of sample 2. The test statistic is the maximum difference between the two empirical CDFs:

$$D_{m,n} = \sup_x |F_m(x) - G_n(x)|$$

For computational purposes:

$$D_{m,n} = \max_{1\le i\le N} |F_m(z_i) - G_n(z_i)|$$

where $z_1, \ldots, z_N$ are the combined ordered observations.

5.2 Distribution Under Null Hypothesis

Under H_0 that the two distributions are identical, the distribution of $D_{m,n}$ depends only on the sample sizes. For large samples:

$$\sqrt{\frac{mn}{m+n}} D_{m,n} \xrightarrow{d} K$$

where K follows the Kolmogorov distribution.

Critical values are available in tables. For m, n large, approximate critical value at $\alpha = 0.05$ is:

$$D_{m,n,0.05} \approx 1.36\sqrt{\frac{m+n}{mn}}$$

Example 5.1 (Kolmogorov-Smirnov Two-Sample Test)**.** Using the pain relief data, test whether the distributions of pain relief times are identical for the two drugs.

Solution:

Step 1: Combine and order the data, compute empirical CDFs

Value	From	$F_m(x)$	$G_n(x)$	$\lvert F_m - G_n \rvert$
18	B	$0/8 = 0.000$	$1/9 = 0.111$	0.111
19	B	$0/8 = 0.000$	$3/9 = 0.333$	0.333
19	B	$0/8 = 0.000$	$3/9 = 0.333$	0.333
20	A	$1/8 = 0.125$	$3/9 = 0.333$	0.208
20	B	$1/8 = 0.125$	$4/9 = 0.444$	0.319
20	B	$1/8 = 0.125$	$5/9 = 0.556$	0.431
21	B	$1/8 = 0.125$	$6/9 = 0.667$	0.542
22	A	$2/8 = 0.250$	$6/9 = 0.667$	0.417
22	B	$2/8 = 0.250$	$7/9 = 0.778$	0.528
23	A	$3/8 = 0.375$	$7/9 = 0.778$	0.403
23	B	$3/8 = 0.375$	$8/9 = 0.889$	0.514
24	A	$4/8 = 0.500$	$8/9 = 0.889$	0.389
24	B	$4/8 = 0.500$	$9/9 = 1.000$	0.500
25	A	$5/8 = 0.625$	$9/9 = 1.000$	0.375
26	A	$6/8 = 0.750$	$9/9 = 1.000$	0.250
27	A	$7/8 = 0.875$	$9/9 = 1.000$	0.125
28	A	$8/8 = 1.000$	$9/9 = 1.000$	0.000

Step 2: Find maximum absolute difference

$$D_{m,n} = \max |F_m - G_n| = 0.542 \quad \text{(at value 21)}$$

Step 3: Compare to critical value For $m = 8$, $n = 9$, the critical value at $\alpha = 0.05$ from tables is approximately 0.455.

Computation of Critical Values for the Two–Sample Kolmogorov–Smirnov Test. In the two–sample Kolmogorov–Smirnov (KS) test, the test statistic is defined as

$$D_{m,n} = \sup_x |F_m(x) - G_n(x)|,$$

where $F_m(x)$ and $G_n(x)$ are the empirical distribution functions of two samples of sizes m and n, respectively. For large sample sizes, the critical value can be approximated using the asymptotic formula

$$D_\alpha \approx c(\alpha)\sqrt{\frac{m+n}{mn}},$$

where $c(\alpha)$ is a constant depending on the significance level α (for example, $c(0.05) = 1.36$). This approximation is derived from the limiting distribution of the KS statistic and is commonly used when both m and n are sufficiently large (typically $m, n \geq 20$). The advantage of this formula is that it allows the critical value to be computed directly without consulting statistical tables.

For small samples, however, the asymptotic approximation can be inaccurate because the exact sampling distribution of $D_{m,n}$ deviates from its limiting form. In such cases, it is recommended to use tabulated critical values obtained from the exact finite–sample distribution of the KS statistic. These tables provide more accurate thresholds for hypothesis testing when the sample sizes are small or moderate.

Application to the Case $m = 8$ and $n = 9$. In the present case, the sample sizes are $m = 8$ and $n = 9$, which are relatively small. Using the asymptotic approximation with $\alpha = 0.05$ gives

$$D_{0.05} \approx 1.36\sqrt{\frac{8+9}{8 \times 9}} \approx 0.661.$$

However, since both sample sizes are small, the more appropriate approach is to consult the exact KS critical value table. From the table, the critical value for $m = 8$ and $n = 9$ at the 5% significance level is approximately

$$D_{0.05}(8, 9) \approx 0.455.$$

Therefore, in this situation the tabulated value should be used for hypothesis testing, and the null hypothesis should be rejected if the observed statistic satisfies

$$D_{m,n} > 0.455.$$

Since $0.542 > 0.455$, we reject H_0.

Step 4: Conclusion The distributions differ at 5% level.

6 Permutation Tests for Two Samples

Permutation tests (also called randomization tests) provide an exact nonparametric approach by randomly reassigning treatment labels.

Basic Idea

Under H_0 that the two distributions are identical, the group labels are arbitrary. We can:

1. Compute a test statistic (e.g., difference in means, difference in medians, t-statistic)
2. Randomly permute the group labels many times
3. Compute the test statistic for each permutation
4. The p-value is the proportion of permutations with test statistic as extreme as observed

Advantages

- Exact test (no approximations)
- Can use any test statistic
- No distributional assumptions

Disadvantages

- Computationally intensive (but modern computers handle this easily)
- Number of permutations grows rapidly with sample size

Example 6.1 (Permutation Test)**.** Using the pain relief data, perform a permutation test for the difference in means.

Solution:

Step 1: Observed data and test statistic Drug A: 20, 25, 22, 28, 24, 26, 23, 27 ($\bar{x}_A = 24.375$) Drug B: 19, 21, 24, 18, 22, 20, 23, 19, 20 ($\bar{x}_B = 20.667$)

Observed difference: $d_{obs} = \bar{x}_A - \bar{x}_B = 3.708$

Step 2: Combine all 17 observations: 20, 25, 22, 28, 24, 26, 23, 27, 19, 21, 24, 18, 22, 20, 23, 19, 20

Step 3: Randomly assign 8 observations to Group A and 9 to Group B, compute difference in means. Repeat many times.

For illustration, consider one permutation: Random assignment: Group A gets: 19, 20, 21, 22, 23, 24, 25, 26 (mean = 22.5) Group B gets: 18, 19, 20, 20, 22, 23, 24, 27, 28 (mean = 22.33) Difference = 0.17

After 10,000 permutations, suppose 247 permutations give a difference ≥ 3.708 in absolute value.

Two-sided p-value = 247/10000 = 0.0247

Step 4: Conclusion Since p-value = 0.0247 < 0.05, we reject H_0. There is significant evidence that the means differ.

7 Wald-Wolfowitz Runs Test

The Wald-Wolfowitz runs test (also called the runs test for two samples) tests whether two samples come from the same population by examining the ordering of observations.

Assumptions

- Two independent random samples from continuous distributions
- Test is sensitive to any difference between distributions (location, scale, shape)

7.1 Test Procedure

1. Combine the two samples and order all observations
2. Replace each observation with a label indicating its source (e.g., A or B)
3. Count the number of runs R, where a run is a consecutive sequence of identical labels
4. Under H_0, the labels are randomly arranged, so the distribution of R is known

7.2 Distribution Under Null Hypothesis

For m observations from sample A and n from sample B, under H_0:

$$\mathbb{E}[R] = \frac{2mn}{m+n} + 1$$

$$\text{Var}(R) = \frac{2mn(2mn - m - n)}{(m+n)^2(m+n-1)}$$

For large samples, R is approximately normal.

Too few runs suggests clustering (samples are separated), too many runs suggests alternating pattern (systematic difference).

Example 7.1 (Wald-Wolfowitz Runs Test)**.** Using the pain relief data, test whether the two samples come from the same population.

Solution:

Step 1: Order all observations and label by group:

Ordered values with labels: 18(B), 19(B), 19(B), 20(A), 20(B), 20(B), 21(B), 22(A), 22(B), 23(A), 23(B), 24(A), 24(B), 25(A), 26(A), 27(A), 28(A)

Step 2: Identify runs (sequences of same label):

Sequence	Run
B B B	Run 1
A	Run 2
B B B	Run 3
A	Run 4
B	Run 5
A	Run 6
B	Run 7
A	Run 8
B	Run 9
A A A A A	Run 10

Number of runs $R = 10$

Step 3: Expected value and variance under H_0:

$$\mathbb{E}[R] = \frac{2mn}{m+n} + 1 = \frac{2 \times 8 \times 9}{17} + 1 = \frac{144}{17} + 1 = 8.47 + 1 = 9.47$$

$$\mathrm{Var}(R) = \frac{2mn(2mn - m - n)}{(m+n)^2(m+n-1)} = \frac{2 \times 72(144 - 17)}{17^2 \times 16} = \frac{144 \times 127}{289 \times 16}$$

$$= \frac{18288}{4624} = 3.95$$

$$SD = \sqrt{3.95} = 1.99$$

Step 4: Normal approximation

$$Z = \frac{R - \mathbb{E}[R]}{SD} = \frac{10 - 9.47}{1.99} = \frac{0.53}{1.99} = 0.27$$

Two-sided p-value $\approx 2 \times \mathbb{P}(Z > 0.27) = 2 \times 0.3936 = 0.7872$

Step 5: Conclusion Since p-value $= 0.787 > 0.05$, we fail to reject H_0. There is insufficient evidence that the samples come from different populations. Note that this test is less powerful than the Wilcoxon test for detecting location shifts.

8 Power and Efficiency Considerations

8.1 Asymptotic Relative Efficiency (ARE)

The ARE of the Wilcoxon rank-sum test relative to the two-sample t-test depends on the underlying distribution:

Distribution	**ARE(Wilcoxon vs. t)**
Normal	$3/\pi \approx 0.955$
Logistic	$\pi^2/9 \approx 1.097$
Laplace (Double Exponential)	1.5
Uniform	1.0
Exponential	3.0
Cauchy	∞

8.2 Comparison of Tests

- **Wilcoxon/Mann-Whitney:** Most powerful for location shifts when distributions have similar shapes; good all-purpose test
- **Median test:** Less powerful than Wilcoxon but robust; useful when data are highly skewed or have outliers
- **Kolmogorov-Smirnov:** Sensitive to all types of differences, not just location; less powerful for pure location shifts
- **Runs test:** Sensitive to general differences but low power; useful as a test of randomness
- **Permutation tests:** Exact, flexible, can be tailored to specific alternatives

8.3 Choosing a Test

- If interested primarily in location shift and distributions have similar shapes: Use Wilcoxon
- If interested in any difference between distributions: Use Kolmogorov-Smirnov
- If sample sizes are very small or exact inference needed: Use permutation test

- If data are highly skewed with many outliers: Consider median test
- If testing for randomness or homogeneity: Consider runs test

9 Exercises

Exercise 9.1 (Wilcoxon Rank-Sum Test)**.** Two methods of teaching reading are compared. Eight students are taught by Method A and seven by Method B. Their reading comprehension scores are:

Method A: 85, 92, 78, 88, 90, 82, 86, 89 Method B: 75, 80, 72, 78, 79, 74, 76

(a) Perform a Wilcoxon rank-sum test at $\alpha = 0.05$ to determine if the methods differ.

(b) Calculate the Mann-Whitney U statistic and verify its relationship to W.

(c) Construct a 95% confidence interval for the shift parameter.

(d) Interpret your results.

Exercise 9.2 (Hodges-Lehmann Estimator)**.** For the data in Exercise 1:

(a) Compute the Hodges-Lehmann estimator of the shift parameter.

(b) Compare it to the difference in sample means.

(c) Which estimator is more robust? Why?

Exercise 9.3 (Median Test)**.** A study compares pain relief times for two drugs:

Drug X: 15, 18, 20, 22, 19, 21, 17, 23, 24, 16 Drug Y: 12, 14, 13, 15, 11, 16, 14, 13, 15, 12

(a) Perform a median test at $\alpha = 0.05$.

(b) Compare the results with a Wilcoxon rank-sum test.

(c) Which test do you think is more appropriate? Why?

Exercise 9.4 (Kolmogorov-Smirnov Test)**.** Using the data from Exercise 3:

(a) Perform a Kolmogorov-Smirnov two-sample test.

(b) What does this test tell you that the Wilcoxon test might miss?

(c) Plot the empirical CDFs for both groups.

Exercise 9.5 (Permutation Test)**.** For the data in Exercise 1:

(a) Perform a permutation test for the difference in means using 10,000 permutations.

(b) Compare the p-value with the Wilcoxon test p-value.

(c) Explain why they might differ.

Exercise 9.6 (Wald-Wolfowitz Runs Test)**.** For the data in Exercise 3:

(a) Perform a runs test for homogeneity.

(b) What does the number of runs tell you about the data?

(c) Compare the conclusion with other tests you've performed.

Exercise 9.7 (Power Comparison)**.** Generate data in R from the following distributions with $m = n = 20$:

(a) Normal(0,1) vs Normal(0.5,1)

(b) Normal(0,1) vs Normal(0,2) (different variances)

(c) Cauchy(0,1) vs Cauchy(0.5,1)

For each case, perform t-test, Wilcoxon test, and Kolmogorov-Smirnov test. Repeat 1000 times and compute empirical power. Summarize your findings in a table. What do you conclude about the relative power of these tests under different alternatives?

Exercise 9.8 (Conceptual Questions)**.** Answer briefly:

(a) Why does the Wilcoxon rank-sum test require the assumption that the distributions have the same shape?

(b) Under what conditions would you prefer the median test over the Wilcoxon test?

(c) What is the advantage of permutation tests over traditional nonparametric tests?

(d) Explain the difference between the null hypotheses of the Wilcoxon test and the Kolmogorov-Smirnov test.

(e) Why might the runs test have low power for detecting location shifts?

A Statistical Tables for Two-Sample Tests

A.1 Critical Values for the Wilcoxon Rank-Sum Test

The following table provides critical values w_α for the Wilcoxon rank-sum test statistic W (sum of ranks in the smaller sample) for one-sided tests at $\alpha = 0.05$ and $\alpha = 0.025$.

Table 1: Critical Values for Wilcoxon Rank-Sum Test (One-Sided)

m	n									
	3	4	5	6	7	8	9	10	11	12
3	6,15	6,18	7,20	7,23	8,25	8,28	9,30	9,33	10,35	10,38
4	6,18	7,21	8,24	9,27	10,30	11,33	12,36	13,39	14,42	15,45
5	7,20	8,24	9,28	11,31	12,35	13,39	15,42	16,46	17,50	19,53
6	7,23	9,27	11,31	12,36	14,40	15,45	17,49	19,53	21,57	22,62
7	8,25	10,30	12,35	14,40	16,45	18,50	20,55	22,60	24,65	26,70
8	8,28	11,33	13,39	15,45	18,50	20,56	23,61	25,67	28,72	30,78
9	9,30	12,36	15,42	17,49	20,55	23,61	26,67	29,73	32,79	35,85
10	9,33	13,39	16,46	19,53	22,60	25,67	29,73	32,80	36,86	39,93
11	10,35	14,42	17,50	21,57	24,65	28,72	32,79	36,86	40,93	44,100
12	10,38	15,45	19,53	22,62	26,70	30,78	35,85	39,93	44,100	48,108

Note: Entries are (lower critical value, upper critical value) for two-sided tests at $\alpha = 0.05$. For one-sided tests at $\alpha = 0.025$, use the same values. Reject H_0 if $W \leq$ lower or $W \geq$ upper.

A.2 Critical Values for the Kolmogorov-Smirnov Two-Sample Test

The following table provides critical values $D_{m,n,\alpha}$ for the Kolmogorov-Smirnov two-sample test.

Table 2: Critical Values for Kolmogorov-Smirnov Two-Sample Test

$m = n$	Significance Level α				
	0.20	0.15	0.10	0.05	0.01
3	2/3	2/3	2/3	2/3	-
4	3/4	3/4	3/4	3/4	-
5	3/5	3/5	4/5	4/5	4/5

Table 2 – continued from previous page

$m = n$	Significance Level α				
	0.20	0.15	0.10	0.05	0.01
6	4/6	4/6	4/6	5/6	5/6
7	4/7	5/7	5/7	5/7	6/7
8	4/8	5/8	5/8	5/8	6/8
9	5/9	5/9	5/9	6/9	7/9
10	5/10	5/10	6/10	6/10	7/10
11	5/11	6/11	6/11	7/11	8/11
12	5/12	6/12	6/12	7/12	8/12
13	6/13	6/13	7/13	7/13	9/13
14	6/14	6/14	7/14	8/14	9/14
15	6/15	7/15	7/15	8/15	9/15
16	6/16	7/16	8/16	8/16	10/16
17	7/17	7/17	8/17	9/17	10/17
18	7/18	8/18	8/18	9/18	10/18
19	7/19	8/19	9/19	9/19	11/19
20	7/20	8/20	9/20	10/20	11/20

Note: For unequal sample sizes, use approximation $D_{m,n,\alpha} \approx c(\alpha)\sqrt{\frac{m+n}{mn}}$ where $c(0.05) = 1.36$, $c(0.01) = 1.63$.

A.3 Critical Values for the Runs Test

The following table provides critical values for the number of runs R in the Wald-Wolfowitz runs test for two samples at $\alpha = 0.05$ (two-sided).

Table 3: Critical Values for Runs Test ($\alpha = 0.05$, Two-Sided)

m	n								
	2	3	4	5	6	7	8	9	10
2	2,5	2,6	2,7	2,8	2,9	2,10	2,11	2,12	2,13
3	2,6	2,7	2,8	3,9	3,10	3,11	3,12	3,13	3,14
4	2,7	2,8	2,9	3,10	3,11	4,12	4,13	4,14	4,15
5	2,8	3,9	3,10	3,11	4,12	4,13	5,14	5,15	5,16
6	2,9	3,10	3,11	4,12	4,13	5,14	5,15	6,16	6,17
7	2,10	3,11	4,12	4,13	5,14	5,15	6,16	6,17	7,18
8	2,11	3,12	4,13	5,14	5,15	6,16	6,17	7,18	7,19

Table 3 – continued from previous page

m	n								
	2	3	4	5	6	7	8	9	10
9	2,12	3,13	4,14	5,15	6,16	6,17	7,18	7,19	8,20
10	2,13	3,14	4,15	5,16	6,17	7,18	7,19	8,20	8,21

Note: Entries are (lower critical value, upper critical value). Reject H_0 if $R \leq$ lower or $R \geq$ upper.

A.4 Chi-Square Distribution Critical Values

For the median test, we need critical values from the chi-square distribution with 1 degree of freedom.

α	$\chi^2_{1,\alpha}$
0.20	1.642
0.15	2.072
0.10	2.706
0.05	3.841
0.025	5.024
0.01	6.635
0.005	7.879

Table 4: Standard Normal Distribution Table: $P(Z \leq z)$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

Table 5: Critical Values of the Chi-Square Distribution

df	0.995	0.99	0.975	0.95	0.90	0.05
1	0.000	0.000	0.001	0.004	0.016	3.841
2	0.010	0.020	0.051	0.103	0.211	5.991
3	0.072	0.115	0.216	0.352	0.584	7.815
4	0.207	0.297	0.484	0.711	1.064	9.488
5	0.412	0.554	0.831	1.145	1.610	11.070
6	0.676	0.872	1.237	1.635	2.204	12.592
7	0.989	1.239	1.690	2.167	2.833	14.067
8	1.344	1.646	2.180	2.733	3.490	15.507
9	1.735	2.088	2.700	3.325	4.168	16.919
10	2.156	2.558	3.247	3.940	4.865	18.307
11	2.603	3.053	3.816	4.575	5.578	19.675
12	3.074	3.571	4.404	5.226	6.304	21.026
13	3.565	4.107	5.009	5.892	7.042	22.362
14	4.075	4.660	5.629	6.571	7.790	23.685
15	4.601	5.229	6.262	7.261	8.547	24.996

Table 6: Two–Sample Kolmogorov–Smirnov Critical Values ($\alpha = 0.05$)

$m \backslash n$	4	5	6	7	8	9	10	11	12
4	0.667	0.640	0.617	0.598	0.582	0.568	0.555	0.544	0.534
5	0.640	0.608	0.583	0.563	0.546	0.531	0.518	0.506	0.495
6	0.617	0.583	0.556	0.534	0.516	0.500	0.486	0.474	0.463
7	0.598	0.563	0.534	0.511	0.492	0.475	0.461	0.448	0.437
8	0.582	0.546	0.516	0.492	0.472	0.455	0.440	0.427	0.416
9	0.568	0.531	0.500	0.475	0.455	0.437	0.422	0.409	0.398
10	0.555	0.518	0.486	0.461	0.440	0.422	0.407	0.394	0.383

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